

Data Analyst III – Advanced SQL

ENTRY LEVEL



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PROFESSIONAL BACKGROUN

As a Certified Scrum Master with over 10 years of professional experience with a background in leading high-performance teams of 8-20 people across technology, business, and design departments. Expert in prioritizing tasks and optimizing workflows for organizational impact. Team-oriented professional with practical experience overseeing business metrics to remove impediments and ensure profitable product delivery. I am currently working as a University Registrar where I create ad-hoc reports on a weekly basis. My goal to attend Entry Level's Data Analyst course is to become a data whiz who can create lasting visualizations that capture the stories of profitable businesses. This portfolio report will further my proficiency with Sequenced Query Languages. I look forward to furthering my skills with SQL and Data Analytics.

UNIVERSITY STUDENT ANALYSIS

Developing a Market Analysis for unique university student trends provides educators and program directors a better understanding of the size and specific segments of a market, as well as an evaluation to determine if the target market will support the business' growth plans. Market Analysis is a critical part of any business plan created, whether to inform the business or to communicate to potential investors the size of the opportunity.

This analysis documents an overview of the higher education industry that your business will participate in. As you narrow this sector down to the ideal customer based on your business strategy, you will define your target market. A detailed description and sizing of the target market will help the reader understand the market value you are pursuing (the number of potential customers multiplied by the average revenue for your product or service).

In defining the target market, you will identify key elements such as geographic location, demographics, your target market's needs and how these needs are being met currently. If there are any direct competitors, these should be compared to how your offering will solve it in the future.

Industry Type: Post-Secondary Education at the university level.

- **Industry Category:** Students likely to attend university based on personal and family details.
- **Industry Characteristics:** We will examine if region, ethnicity, gender, and areas of urbanization affect wages and motivate college enrollment.
- **Trends:** Women have outpaced university enrollment over their male counterparts.
- **Stability:** Historically, urban areas have seen students steadily enroll in university while students outside of urbanized areas have been increasing their enrollments to catch up with their urban counterparts.

ROOT CAUSE ANALYSIS

Students from lower income, urban areas typically in the East tend to attend university at a higher rate than students of other demographics. The reason for this can be because bigger cities require individuals with more specialized training. This report will seek to understand which students are enrolling into universities and what are their backgrounds. This will help college directors better understand which

students would be best targets for growth and expansion. Also, university degrees have long been thought to alleviate the income gap between diverse demographics. This report will examine this theory for areas of strength and weakness.

ANALYSIS

Method:

The method used to obtain the data used in this report was SQL and Excel. Four datasets were downloaded into CSV files and uploaded using Excel. These Excel spreadsheets were imported into the SQL software platform where they were joined using the following query code:

```
SELECT *
FROM student_personal_details
JOIN student_family_details on student_personal_details.ID =
student_family_details.id
Join student_academic_info on student_family_details.id =
student_academic_info.id
Join county_info ON student_academic_info.id = county_info.id
```

To limit the data size, the following query was used to look at the top 20 primary key to identify student demographical information of gender, ethnicity, home ownership, urban, income, unemployment rating, wage, and the region where they reside:

```
SELECT gender, ethnicity, home, urban, income, unemp, wage,
region
FROM student_personal_details
JOIN student_family_details on student_personal_details.ID =
student_family_details.id
Join student_academic_info on student_family_details.id =
student_academic_info.id
Join county_info ON student_academic_info.id = county_info.id
limit 20
```

To understand which students were engaging in universities, the following query was used to identify academic involvement based on gender, ethnicity:

```

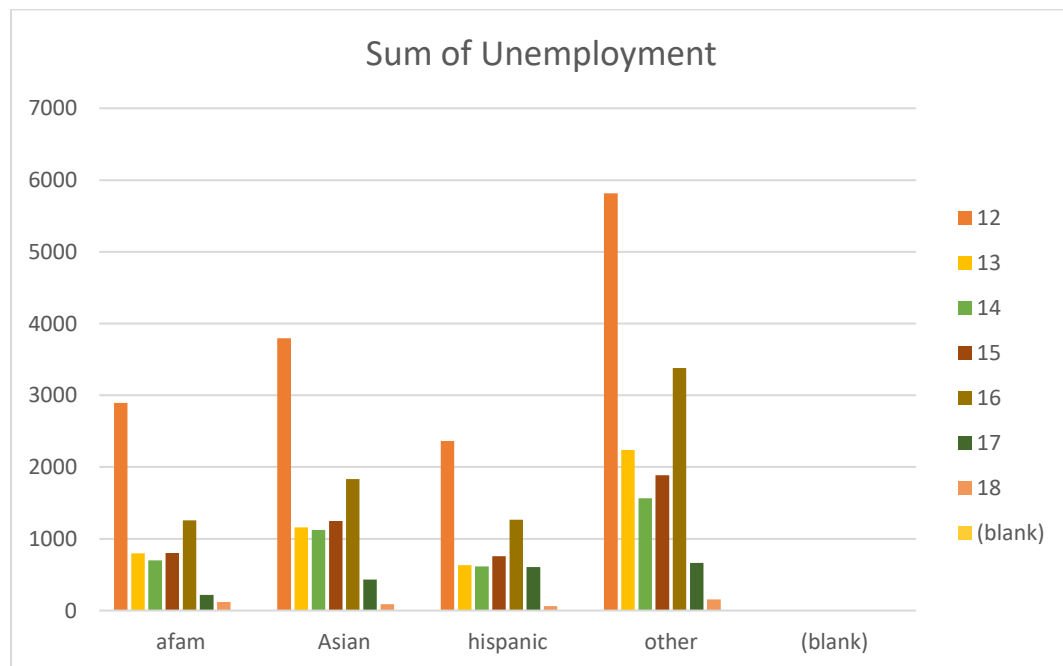
SELECT gender, ethnicity, academic_score, education
FROM student_personal_details
Join student_academic_info on student_personal_details.ID
Join county_info ON student_academic_info.id = county_info.id

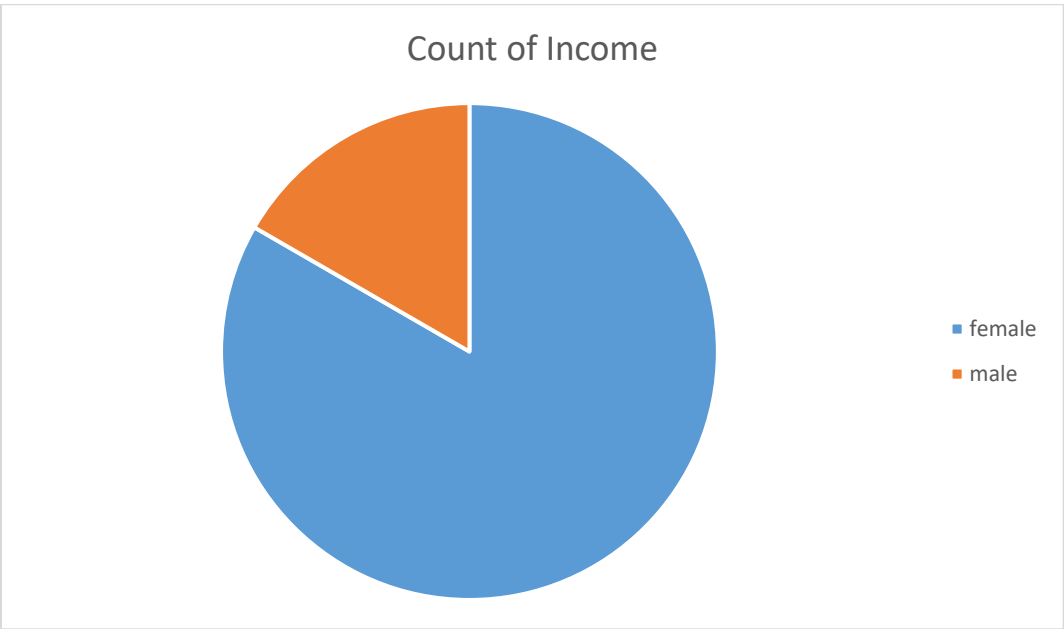
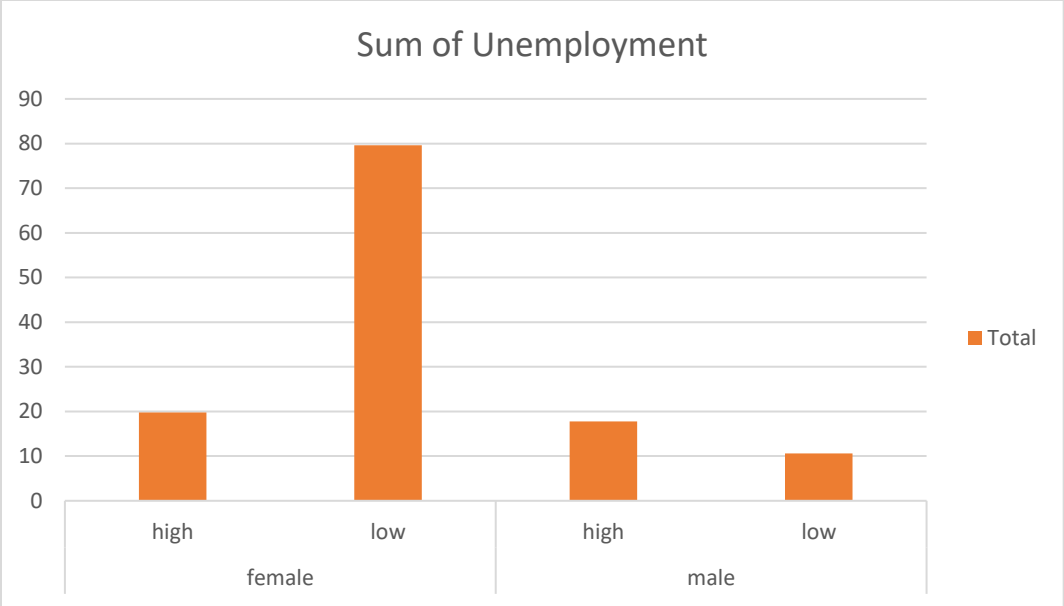
```

Insights:

ID	gender	DOB	ethnicity	academic	county_id	family_deid	fcollege	mcollege	home	urban	income	id	academic	student_t	education id	unemp	
6	male	25/5/1992	other	691	3876	3980	6 yes	yes	no	no	high	6	38.22	65416	13	6	8.9
6	male	25/5/1992	other	691	3876	3980	6 yes	yes	no	no	high	6	44.74	74070	12	6	8.9
7	female	22/4/1987	Asian	2082	3595	1750	7 no	no	yes	no	high	7	61.05	69638	16	7	4.4
7	female	22/4/1987	Asian	2082	3595	1750	7 no	no	yes	no	high	7	61.05	69638	16	7	5.5
7	female	22/4/1987	Asian	2082	3595	1750	7 no	no	yes	no	high	7	62.81	111624	16	7	4.4
7	female	22/4/1987	Asian	2082	3595	1750	7 no	no	yes	no	high	7	62.81	111624	16	7	5.5
7	female	22/4/1987	Asian	2082	3595	1750	7 no	no	yes	yes	low	7	61.05	69638	16	7	4.4
7	female	22/4/1987	Asian	2082	3595	1750	7 no	no	yes	yes	low	7	61.05	69638	16	7	5.5
7	female	22/4/1987	Asian	2082	3595	1750	7 no	no	yes	yes	low	7	62.81	111624	16	7	4.4
7	female	22/4/1987	Asian	2082	3595	1750	7 no	no	yes	yes	low	7	62.81	111624	16	7	5.5
10	female	6/9/1993	other	1268	3448	3332	10 yes	yes	yes	no	low	10	41.47	61328	16	10	5.5
10	female	6/9/1993	other	1268	3448	3332	10 yes	yes	yes	no	low	10	41.47	61328	16	10	6.3
19	female	22/3/1998	other	4501	492	3256	19 no	no	yes	no	low	19	37.77	61101	15	19	8
19	female	22/3/1998	other	4501	492	3256	19 no	no	yes	no	low	19	62.92	50340	16	19	8
19	female	22/3/1998	other	4501	492	3256	19 no	no	yes	no	low	19	63.92	86750	12	19	8
19	female	22/3/1998	other	4501	492	3256	19 no	yes	no	no	low	19	37.77	61101	15	19	8
19	female	22/3/1998	other	4501	492	3256	19 no	yes	no	no	low	19	62.92	50340	16	19	8
19	female	22/3/1998	other	4501	492	3256	19 no	yes	no	no	low	19	63.92	86750	12	19	8
23	male	17/1/1996	other	1502	1732	1996	23 no	no	yes	no	low	23	43.64	82942	12	23	5.3
23	male	17/1/1996	other	1502	1732	1996	23 no	no	yes	no	low	23	43.91	106366	16	23	5.3
23	male	17/1/1996	other	1502	1732	1996	23 no	no	yes	no	low	23	44.48	61671	14	23	5.3
23	male	17/1/1996	other	1502	1732	1996	23 no	no	yes	no	low	23	66.31	103539	12	23	5.3
23	male	17/1/1996	other	1502	1732	1996	23 yes	no	yes	yes	high	23	43.64	82942	12	23	5.3
23	male	17/1/1996	other	1502	1732	1996	23 yes	no	yes	yes	high	23	43.91	106366	16	23	5.3
23	male	17/1/1996	other	1502	1732	1996	23 yes	no	yes	yes	high	23	44.48	61671	14	23	5.3
23	male	17/1/1996	other	1502	1732	1996	23 yes	no	yes	yes	high	23	66.31	103539	12	23	5.3
34	male	14/1/1983	other	705	2721	3902	34 no	no	no	no	low	34	59.11	71956	16	34	4.3
34	male	14/1/1983	other	705	2721	3902	34 no	no	no	no	low	34	59.11	71956	16	34	4.3
34	male	14/1/1983	other	705	2721	3902	34 no	no	no	no	low	34	59.11	71956	16	34	6.3
34	male	14/1/1983	other	705	2721	3902	34 no	no	no	no	low	34	59.11	71956	16	34	8.7
34	male	14/1/1983	other	705	2721	3902	34 no	no	yes	no	low	34	59.11	71956	16	34	4.3
34	male	14/1/1983	other	705	2721	3902	34 no	no	yes	no	low	34	59.11	71956	16	34	4.3
34	male	14/1/1983	other	705	2721	3902	34 no	no	yes	no	low	34	59.11	71956	16	34	6.3
34	male	14/1/1983	other	705	2721	3902	34 no	no	yes	no	low	34	59.11	71956	16	34	8.7
40	female	19/1/1989	other	1685	3424	2167	40 no	no	no	yes	low	40	47.25	77126	16	40	8.5
40	female	19/1/1989	other	1685	3424	2167	40 no	no	no	yes	low	40	48.41	102531	12	40	8.5
40	female	19/1/1989	other	1685	3424	2167	40 no	no	yes	no	low	40	47.25	77126	16	40	8.5
40	female	19/1/1989	other	1685	3424	2167	40 no	no	yes	no	low	40	48.41	102531	12	40	8.5
40	female	19/1/1989	other	1685	3424	2167	40 no	no	yes	no	low	40	47.25	77126	16	40	8.5
40	female	19/1/1989	other	1685	3424	2167	40 no	no	yes	no	low	40	48.41	102531	12	40	8.5
41	male	24/2/1998	Asian	4178	1781	3023	41 no	no	yes	yes	low	41	59.01	63885	16	41	7.5
42	female	13/7/1991	other	2047	228	4390	42 no	no	no	no	high	42	57.78	55343	13	42	11.2
42	female	13/7/1991	other	2047	228	4390	42 no	no	no	no	high	42	64.86	114961	16	42	11.2
46	female	1/3/1985	Asian	1493	33	3349	46 no	no	no	no	low	46	34.42	92088	12	46	9.2
47	female	9/7/1998	afam	3669	3413	4422	47 no	no	yes	no	low	47	55.73	78645	12	47	6.4
47	female	9/7/1998	afam	3669	3413	4422	47 no	no	yes	no	low	47	55.73	78645	12	47	7.8
47	female	9/7/1998	afam	3669	3413	4422	47 no	no	yes	no	low	47	55.73	78645	12	47	10.8
47	female	9/7/1998	afam	3669	3413	4422	47 no	no	yes	no	low	47	55.73	78645	12	47	6.4
47	female	9/7/1998	afam	3669	3413	4422	47 no	no	yes	no	low	47	55.73	78645	12	47	7.8
47	female	9/7/1998	afam	3669	3413	4422	47 no	no	yes	no	low	47	55.73	78645	12	47	10.8
47	female	9/7/1998	afam	3669	3413	4422	47 no	no	yes	yes	low	47	55.73	78645	12	47	6.4
47	female	9/7/1998	afam	3669	3413	4422	47 no	no	yes	yes	low	47	55.73	78645	12	47	7.8
47	female	9/7/1998	afam	3669	3413	4422	47 no	no	yes	yes	low	47	55.73	78645	12	47	10.8
47	female	9/7/1998	afam	3669	3413	4422	47 yes	yes	yes	no	high	47	55.73	78645	12	47	6.4
47	female	9/7/1998	afam	3669	3413	4422	47 yes	yes	yes	no	high	47	55.73	78645	12	47	7.8
47	female	9/7/1998	afam	3669	3413	4422	47 yes	yes	yes	no	high	47	55.73	78645	12	47	10.8
49	male	7/7/1990	hispanic	2722	1990	2448	49 no	no	yes	no	low	49	65.07	78276	17	49	9.6
57	female	24/6/1981	other	3959	3115	4651	57 no	yes	no	yes	low	57	41.08	87162	15	57	4.7
57	female	24/6/1981	other	3959	3115	4651	57 no	yes	no	yes	low	57	41.08	87162	15	57	8
57	female	24/6/1981	other	3959	3115	4651	57 yes	no	yes	yes	high	57	41.08	87162	15	57	4.7

gender	ethnicity	home	urban	income	unemp	wage	region
male	other	no	no	high	8.9	9.96	east
male	other	no	no	high	8.9	9.96	east
female	Asian	yes	no	high	4.4	9.64	east
female	Asian	yes	no	high	5.5	7.18	other
female	Asian	yes	no	high	4.4	9.64	east
female	Asian	yes	no	high	5.5	7.18	other
female	Asian	yes	yes	low	4.4	9.64	east
female	Asian	yes	yes	low	5.5	7.18	other
female	Asian	yes	yes	low	4.4	9.64	east
female	Asian	yes	yes	low	5.5	7.18	other
female	other	yes	no	low	5.5	8.09	other
female	other	yes	no	low	6.3	9.68	other
female	other	no	no	low	8	10.28	east
female	other	no	no	low	8	10.28	east
female	other	no	no	low	8	10.28	east
female	other	yes	no	low	8	10.28	east
female	other	yes	no	low	8	10.28	east
female	other	yes	no	low	8	10.28	east
male	other	yes	no	low	5.3	9.64	other
male	other	yes	no	low	5.3	9.64	other





FINDINGS

The findings demonstrate that a lack of college degree impacts wages and significantly impacts how length of unemployment for women more than men. Due to this significant disparity, women are enrolling in university at a higher rate than their male counterparts. Students of African American, Asian, and Hispanic ethnicities seem to be keeping up the pace with university enrollments while individuals listed as other are not enrolling as much. Unemployment also appears to be higher for those within the other group as

opposed to students who identify as minority students. These findings are indicating a trend of students from diverse backgrounds are seeking increased income and employability through post-secondary education. Of these students, female students more than male students are particularly invested in obtaining university degrees possibly because they suffer the highest rate of low income and unemployment.

CONCLUSION

Based on the findings, women are more likely than men to enroll into university due to having lower income. The data further demonstrates that Asian women with lower income are more motivated to attend university to possibly better their wage outlook. The findings also demonstrate that is a wage gap present between men and women especially for lower wage-earning women. Overall, universities in all regions should aim to attract more lower income Asian students by identifying their needs and industries of interest to increase their enrollments with these prospective students. From this report we can see that males are not enrolling into university as much as women and that may be due in part by their income not being impacted by external factors such as region, gender, living in urban areas or homeownership. University directors and administrators should specifically target graduates who live in non-urban areas and who come from lower-income backgrounds.